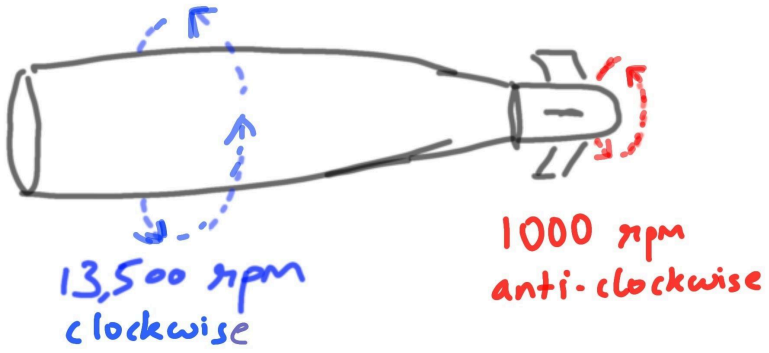
PGK: Ideation and Methodology

Abstract—This paper presents the ideation and methodology for PGK. It describes generating power from the relative rotation of two parts turning in opposite directions (13,500 rpm clockwise and 1000 rpm anticlockwise, giving a relative speed of approximately 14,500 rpm), a four-fin arrangement in which rotation speed decides whether lift is distributed or directed, three operating modes (no brakes at high rpm for power generation only; low rpm with initial braking, in which the sensors start; and braking with the lift vector juggled), the drop in relative rotation during braking (from 14,500 to 13,800 rpm), the juggling manoeuvre method by which brakes are engaged and released to steer the lift vector, and a method of accumulating time spent in the lift direction, which is the basis for the simulation.

Keywords—relative rotation, power generation, lift vector, fin pairs, braking, juggling manoeuvre, simulation

I. POWER GENERATION FROM RELATIVE ROTATION

Two rotating parts turn in opposite directions (Fig. 1). The main body turns clockwise at 13,500 rpm and the finned section turns anticlockwise at 1000 rpm. Because they rotate against each other, the speed between them (the relative speed) is approximately 14,500 rpm. Electrical generation is derived from this relative speed. The corresponding generation, in kW, remains to be determined.



Clockwise rotation: 13,500 rpm.

Anticlockwise rotation: 1000 rpm.

Relative speed: approximately 14,500 rpm.

Fig. 1. Counter-rotating parts used for power generation.

II. FIN ARRANGEMENT: ROTATIONAL PAIRS AND LIFT PAIRS

The ring carries four fins arranged as two opposite pairs, designated *rotational pairs* and *lift pairs*. The speed of rotation decides how the lift behaves:

- High speed: lift is distributed around the ring, so directed lift = 0.
- Low speed: lift is directed.

III. OPERATING MODES

The system operates in three modes (Fig. 2).

- **Mode 1:** no brakes, high rpm (approximately 1000 rpm). Only power generation takes place and the sensors are off.
- **Mode 2:** low rpm, initial braking (300–400 rpm). The sensors kick in. This mode is important, because it is what allows the sensors to start working. This also incorporates maneuvering while braking.
- **Mode 3:** braking, with the lift vector juggled (0–100 rpm). This is the maneuvering mode.

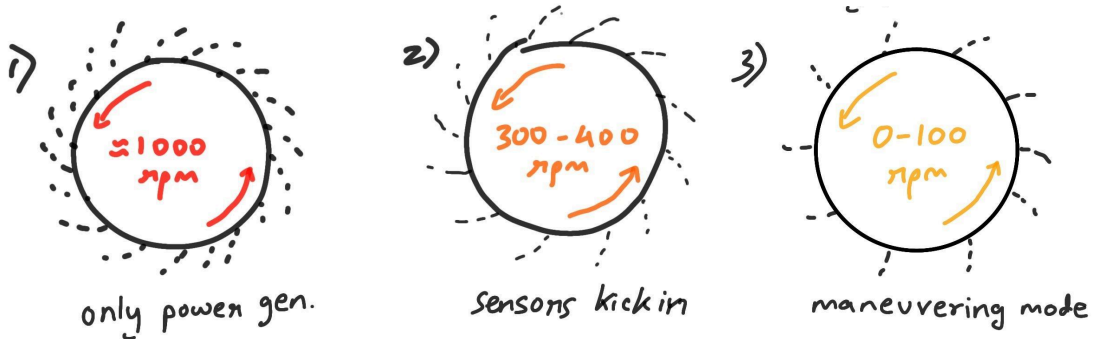


Fig. 2. Operating modes: (1) only power generation, (2) sensors kick in, (3) manoeuvring mode.

Note: (2) also incorporates braking while braking method (explained further below).

A. Manoeuvring approach

An ideal case is distinguished from a realistic plan (Table I).

TABLE I
MANOEUVRING APPROACHES

Case	Description
Ideal manoeuvring	The plant stays stationary to the ground while maneuvering.
Realistic plan	The plant rotates and spends higher amounts of time in the required direction. Stationary operation accounts for misses in juggling the lift vector.

IV. BRAKING AND ITS EFFECT ON ROTATION SPEED

Normal operation is compared with braking in Table II. The outer ring stays at 13,500 rpm. During braking the inner rotor slows from 1000 rpm to 300 rpm. As seen from the alternator, the relative rotation therefore falls from 14,500 rpm to 13,600–13,800 rpm. The effect is that rotation drops from 14,500 to 13,800 rpm. Fig. 3 shows manoeuvring while slowing down, with the braking zone (slower), the brake release zone (faster) and the resulting lift vector.

TABLE II
ROTATION SPEEDS IN NORMAL OPERATION AND BRAKING

Modes	**Outer ring (wrt ground)**	**Inner rotor (wrt ground)**	**Relative (alternator)**
Normal (Mode 1)	13,500 rpm	1000 rpm	14,500 rpm
Braking (Mode 2)	13,500 rpm	300 rpm	13,600-13,800 rpm
Near Zero RPM (Mode3)	13,500 rpm	0-100 rpm	13,500-13,600 rpm

V. METHOD OF ACCUMULATING TIME SPENT (RPM 100-300)

A. Why accumulation is needed

In Mode 3 the lift is directed, but the plant is rotating, so the fins do not point in the required direction all the time. The method of accumulating time spent counts how much time the fins spend in the required direction, and adds this up over many revolutions until the total is large enough to correct the trajectory.

B. Time spent in one revolution

The part of the revolution in which the brakes are engaged is called the braking sector. It spans an angle Δθ. The fraction of one revolution spent in this sector is

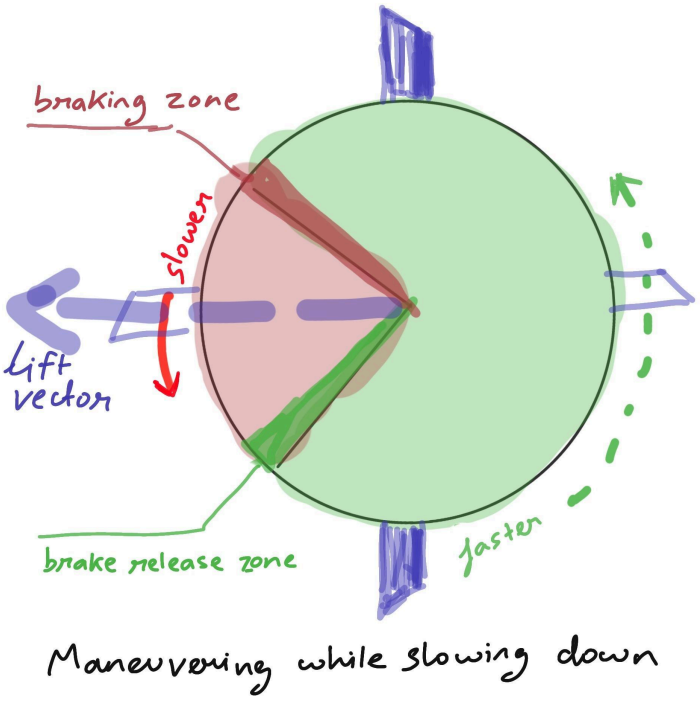
$f = \Delta\theta / 360°$ (1)

At a rotor speed of n rpm, one revolution takes

$T = 60 / n$ seconds (2)

For example, T = 0.6 s at 100 rpm, the top of the Mode 3 range. The time spent in the lift direction during one revolution is therefore

$t_{rev} = f \cdot T$ (3)



C. Time accumulated over many revolutions

Each revolution adds the same small amount of time in the lift direction. Over N revolutions the accumulated time is

$t_{acc} = N \cdot f \cdot T = f \cdot t$ (4)

where $t = N \cdot T$ is the total manoeuvring time. Equation (4) shows that the accumulated time depends on the sector fraction f and the total time t, and not on the rotor speed n. Rotor speed only decides how often the pulses occur (n/60 pulses per second), not how much time is accumulated.*Illustrative example (not design values):* with $\Delta\theta = 90°$, f = 0.25. If the rotor turns at 60 rpm, T = 1 s. Over a 10 s manoeuvre, N = 10 revolutions and $t_{acc} = 10 \times 0.25 \times 1$ s = 2.5 s.

VI. JUGGLING MANOEUVRE METHOD (RPM 0-100)

A. Principle

In Mode 3 the inner rotor, which carries the finned ring, turns at only 0–100 rpm. At this low speed the lift is no longer distributed around the ring but is directed (Section II). The juggling manoeuvre exploits this: instead of holding one fixed brake setting, the alternators alternately engage and release their brakes so that the fins are slowed in one part of the revolution and allowed to speed up in the rest. The fins therefore spend more time in the braking sector than elsewhere, and the lift vector is pushed towards the direction the vehicle needs.

Two regions of the revolution are distinguished (Fig. 3 and Fig. 4):

- **Braking zone (brakes engaged):** the fins move slower, and lift is held in the required direction for longer.
- **Brake release zone (brakes released):** the fins move faster, so little time is spent there and little net lift results.

Because the brakes are engaged and released in turn, the lift vector is "juggled" between the two states shown in Fig. 4 rather than being fixed in one direction.

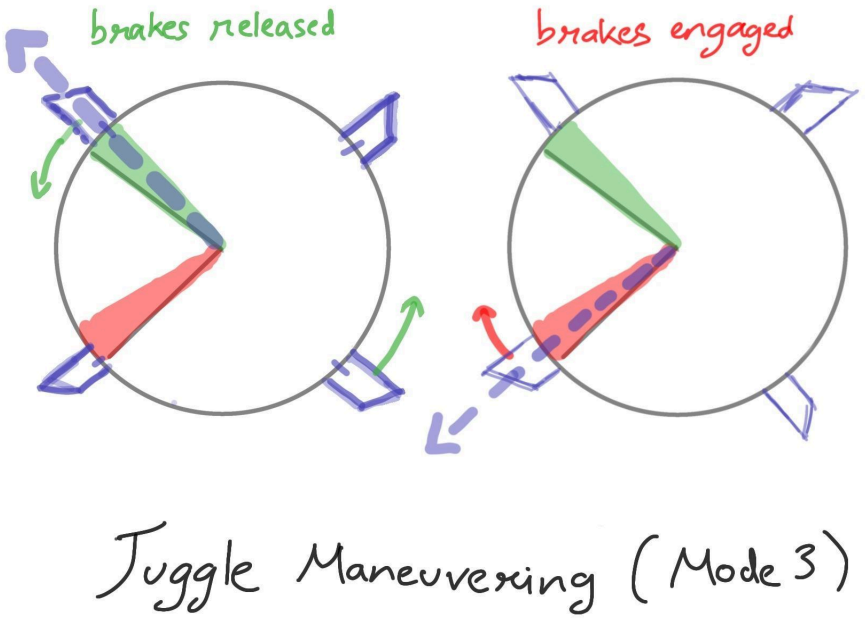


Fig. 4. Juggle manoeuvring (Mode 3): brakes released and brakes engaged.

B. Sequence of operation

1. **Sensors on (Mode 2).** The rotor is slowed to 300–400 rpm by initial braking so that the sensors start working and the required correction direction can be determined.
2. **Enter manoeuvring (Mode 3).** The rotor is slowed further into the 0–100 rpm range, where lift is directed.
3. **Select the braking sector.** The braking sector is placed on the side of the revolution that corresponds to the required lift direction.
4. **Engage brakes** as the fins pass through the braking sector. The fins slow and the lift vector points in the required direction (Fig. 4, brakes engaged).

5. **Release brakes** outside the sector. The fins speed up through the brake release zone (Fig. 4, brakes released).
6. **Repeat every revolution.** The short periods of directed lift add up (Section VI-C) and the vehicle moves from its initial trajectory onto the corrected trajectory (Section VII).
7. **Return to Mode 1 or Mode 2** once the correction is complete, restoring full power generation.

VII. SIMULATION APPROXIMATION AND TRAJECTORY CORRECTION

The effect is cumulative, so the system spends only a small amount of time in the lift direction. This is equal to a pulse in that same direction, which is the quantity used for the simulation. In the approximation, the lift coefficient is $C_L = C_{L2}$ within the braking sector and $C_L = 0$ elsewhere on the circle, which is equivalent to a pulse in C_L (Fig. 5). The resulting correction moves the vehicle from its initial trajectory onto a corrected trajectory (Fig. 6).

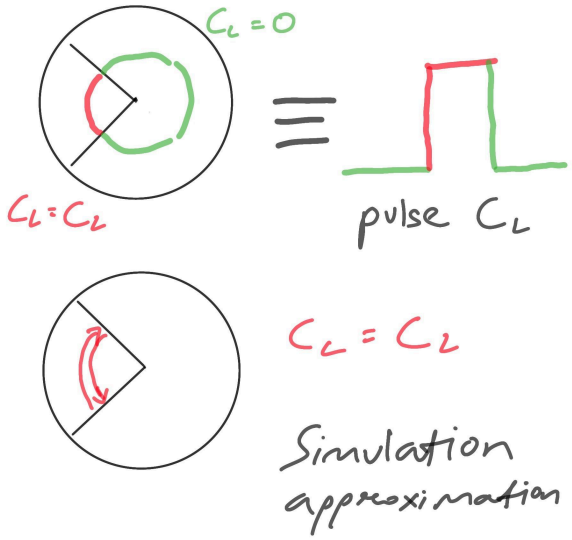


Fig. 5. Simulation approximation: the lift coefficient over a revolution as a pulse in C_L.

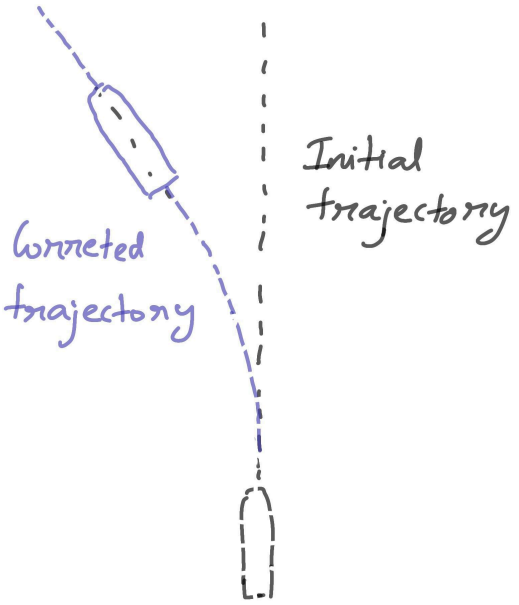


Fig. 6. Initial trajectory and corrected trajectory.

Pulse equivalent and mean lift coefficient

Because the system spends only a small amount of time in the lift direction in each revolution, the effect is equal to a pulse in that same direction (Fig. 5). In the approximation used for the simulation:

- C_L = C_L2 inside the braking sector,
- C_L = 0 elsewhere on the circle.

Over one revolution this is a rectangular pulse of height C_L2 and width f · T, repeated every T seconds. Averaged over a revolution, the mean lift coefficient is

C_L,mean = f · C_L2 (5)

If air density, speed and fin area stay roughly constant during the manoeuvre, the lift force is proportional to C_L. The total effect of the manoeuvre is then proportional to the accumulated pulse area:

effect $\propto$ C_L2 · t_acc = C_L2 · f · t (6)

This is why the effect is cumulative: one revolution gives only a small pulse, and repeating it over many revolutions builds up the correction that moves the vehicle from its initial trajectory onto the corrected trajectory (Fig. 6).

VIII. SUMMARY OF FIGURES

Table IV collects the numerical values quoted in this paper.

TABLE IV
SUMMARY OF QUOTED FIGURES

Quantity	**Value**
Clockwise rotation	13,500 rpm
Anticlockwise rotation (normal)	1000 rpm
Relative speed (normal)	approx. 14,500 rpm
Anticlockwise rotation (braking)	300 rpm
Relative speed (braking)	13,600–13,800 rpm
Rotation after braking	14,500 down to 13,800 rpm
Mode 1 speed (no brakes, high rpm)	approx. 1000 rpm
Mode 2 speed (initial braking, sensors on)	300–400 rpm
Mode 3 speed (braking, manoeuvring)	0–100 rpm